

```
In [ ]:
```

```
In [1]: import pandas as pd
import numpy as np
```

```
In [2]: df = pd.read_csv("./Data/aug_train.csv")
df.head()
```

Out[2]:

	enrollee_id	city	city_development_index	gender	relevent_experience	enrolled_univ
0	8949	city_103	0.920	Male	Has relevent experience	no_enrol
1	29725	city_40	0.776	Male	No relevent experience	no_enrol
2	11561	city_21	0.624	NaN	No relevent experience	Full time c
3	33241	city_115	0.789	NaN	No relevent experience	
4	666	city_162	0.767	Male	Has relevent experience	no_enrol

```
In [3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19158 entries, 0 to 19157
Data columns (total 14 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   enrollee_id                          19158 non-null  int64
1   city                                 19158 non-null  object
2   city_development_index               19158 non-null  float64
3   gender                               14650 non-null  object
4   relevent_experience                  19158 non-null  object
5   enrolled_university                 18772 non-null  object
6   education_level                     18698 non-null  object
7   major_discipline                    16345 non-null  object
8   experience                           19093 non-null  object
9   company_size                        13220 non-null  object
10  company_type                         13018 non-null  object
11  last_new_job                         18735 non-null  object
12  training_hours                       19158 non-null  int64
13  target                               19158 non-null  float64
dtypes: float64(2), int64(2), object(10)
memory usage: 2.0+ MB
```

```
In [4]: df = df.drop("city",axis=1)
df.head()
```

Out[4]:

	enrollee_id	city_development_index	gender	relevent_experience	enrolled_university	ec
0	8949	0.920	Male	Has relevent experience	no_enrollment	
1	29725	0.776	Male	No relevent experience	no_enrollment	
2	11561	0.624	NaN	No relevent experience	Full time course	
3	33241	0.789	NaN	No relevent experience	NaN	
4	666	0.767	Male	Has relevent experience	no_enrollment	

Handling Missing Values

```
In [5]: df.isna().sum()

Out[5]: enrollee_id          0
city_development_index      0
gender          4508
relevent_experience          0
enrolled_university         386
education_level            460
major_discipline           2813
experience                   65
company_size                5938
company_type                6140
last_new_job                423
training_hours              0
target                      0
dtype: int64

In [6]: ##for gender, replace with mode.
df["gender"] = df["gender"].fillna(df["gender"].mode()[0])
df["gender"].head(),df["gender"].isna().sum()

Out[6]: (0    Male
1    Male
2    Male
3    Male
4    Male
Name: gender, dtype: object,
0)

In [7]: df["enrolled_university"].unique()

Out[7]: array(['no_enrollment', 'Full time course', nan, 'Part time course'],
dtype=object)

In [8]: df["enrolled_university"] = df["enrolled_university"].fillna(df["enrolled_universit
df["enrolled_university"].head(),df["enrolled_university"].isna().sum()

Out[8]: (0    no_enrollment
1    no_enrollment
2    Full time course
3    no_enrollment
4    no_enrollment
Name: enrolled_university, dtype: object,
0)

In [9]: df["education_level"].unique()

Out[9]: array(['Graduate', 'Masters', 'High School', nan, 'Phd', 'Primary School'],
dtype=object)

In [10]: df["education_level"] = df["education_level"].fillna(df["education_level"].mode()[0]
df["education_level"].head(),df["education_level"].isna().sum()

Out[10]: (0    Graduate
1    Graduate
2    Graduate
3    Graduate
4    Masters
Name: education_level, dtype: object,
0)

In [11]: df["major_discipline"] = df["major_discipline"].fillna(df["major_discipline"].mode(
df["major_discipline"].head(),df["major_discipline"].isna().sum()
```

```
Out[11]: (0          STEM
         1          STEM
         2          STEM
         3  Business Degree
         4          STEM
         Name: major_discipline, dtype: object,
         0)
```

```
In [12]: df["experience"].unique()
```

```
Out[12]: array(['>20', '15', '5', '<1', '11', '13', '7', '17', '2', '16', '1', '4',
               '10', '14', '18', '19', '12', '3', '6', '9', '8', '20', nan],
              dtype=object)
```

```
In [13]: df["experience"] = df["experience"].replace('>20','21')
df["experience"] = df["experience"].replace('<1','0')
df["experience"].unique()
```

```
Out[13]: array(['21', '15', '5', '0', '11', '13', '7', '17', '2', '16', '1', '4',
               '10', '14', '18', '19', '12', '3', '6', '9', '8', '20', nan],
              dtype=object)
```

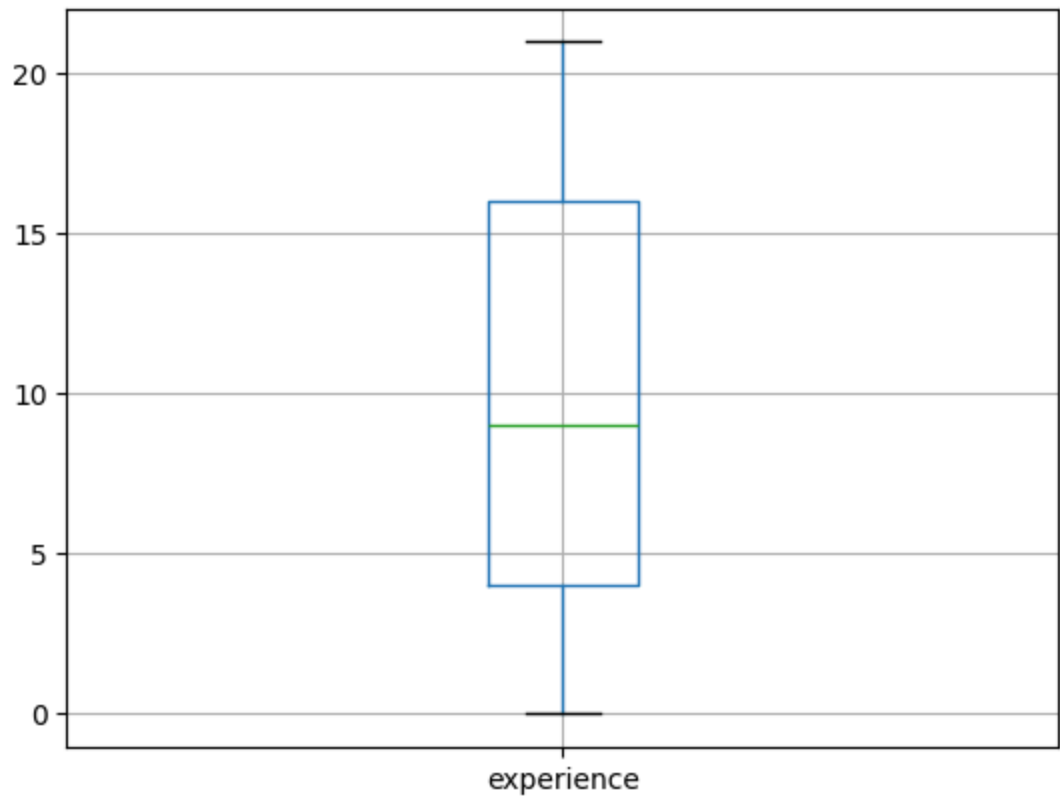
```
In [14]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19158 entries, 0 to 19157
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   enrollee_id           19158 non-null  int64
1   city_development_index 19158 non-null  float64
2   gender                 19158 non-null  object
3   relevent_experience     19158 non-null  object
4   enrolled_university    19158 non-null  object
5   education_level        19158 non-null  object
6   major_discipline       19158 non-null  object
7   experience             19093 non-null  object
8   company_size           13220 non-null  object
9   company_type           13018 non-null  object
10  last_new_job            18735 non-null  object
11  training_hours         19158 non-null  int64
12  target                 19158 non-null  float64
dtypes: float64(2), int64(2), object(9)
memory usage: 1.9+ MB
```

```
In [15]: df["experience"] = df["experience"].astype(float)
df["experience"].unique()
```

```
Out[15]: array([21., 15., 5., 0., 11., 13., 7., 17., 2., 16., 1., 4., 10.,
               14., 18., 19., 12., 3., 6., 9., 8., 20., nan])
```

```
In [16]: import matplotlib.pyplot as plt
temp = pd.DataFrame(df["experience"])
temp.boxplot()
plt.show()
```



```
In [17]: df["experience"] = df["experience"].fillna(df["experience"].mean())
df["experience"].head(),df["experience"].unique()
```

```
Out[17]: (0    21.0
1     15.0
2      5.0
3      0.0
4     21.0
Name: experience, dtype: float64,
array([21.      , 15.      ,  5.      ,  0.      , 11.      ,
       13.      ,  7.      , 17.      ,  2.      , 16.      ,
         1.      ,  4.      , 10.      , 14.      , 18.      ,
       19.      , 12.      ,  3.      ,  6.      ,  9.      ,
         8.      , 20.      , 10.10014141]))
```

```
In [18]: df["company_size"].unique()
```

```
Out[18]: array([nan, '50-99', '<10', '10000+', '5000-9999', '1000-4999', '10/49',
       '100-500', '500-999'], dtype=object)
```

```
In [19]: df["company_size"] = df["company_size"].replace('<10','0-10')
df["company_size"] = df["company_size"].replace('10/49','10-49')
df["company_size"].unique()
```

```
Out[19]: array([nan, '50-99', '0-10', '10000+', '5000-9999', '1000-4999', '10-49',
       '100-500', '500-999'], dtype=object)
```

```
In [20]: df["company_size"] = df["company_size"].fillna(df["company_size"].mode()[0])
df["company_size"].head()
```

```
Out[20]: 0    50-99
1    50-99
2    50-99
3    50-99
4    50-99
Name: company_size, dtype: object
```

```
In [21]: df["company_size"].unique()
```

```
Out[21]: array(['50-99', '0-10', '10000+', '5000-9999', '1000-4999', '10-49',
       '100-500', '500-999'], dtype=object)
```

```
In [22]: df["company_type"].unique()
```

```
Out[22]: array([nan, 'Pvt Ltd', 'Funded Startup', 'Early Stage Startup', 'Other',
       'Public Sector', 'NGO'], dtype=object)
```

```
In [23]: df["company_type"] = df["company_type"].fillna(df["company_type"].mode()[0])
df["company_type"].head(),df["company_type"].isna().sum()
```

Out[23]: (0 Pvt Ltd
1 Pvt Ltd
2 Pvt Ltd
3 Pvt Ltd
4 Funded Startup
Name: company_type, dtype: object,
0)

```
In [24]: df["last_new_job"].unique()
```

Out[24]: array(['1', '>4', 'never', '4', '3', '2', nan], dtype=object)

```
In [25]: df["last_new_job"] = df["last_new_job"].replace('>4','5')
df["last_new_job"] = df["last_new_job"].replace('never','0')
df["last_new_job"] = df["last_new_job"].astype(float)
df["last_new_job"].unique()
```

Out[25]: array([1., 5., 0., 4., 3., 2., nan])

```
In [26]: df["last_new_job"] = df["last_new_job"].fillna(df["last_new_job"].median())
df["last_new_job"].unique()
```

Out[26]: array([1., 5., 0., 4., 3., 2.])

```
In [27]: df["training_hours"].unique()
```

Out[27]: array([36, 47, 83, 52, 8, 24, 18, 46, 123, 32, 108, 23, 26,
106, 7, 132, 68, 50, 48, 65, 13, 22, 148, 72, 40, 141,
 82, 145, 206, 152, 42, 14, 112, 87, 20, 21, 92, 102, 43,
 45, 19, 90, 25, 15, 98, 142, 28, 228, 29, 12, 17, 35,
 4, 136, 27, 74, 86, 75, 332, 140, 182, 172, 33, 34, 150,
160, 3, 2, 210, 101, 59, 260, 131, 109, 70, 51, 60, 164,
290, 133, 76, 156, 120, 100, 39, 55, 49, 6, 125, 326, 198,
 11, 41, 114, 246, 81, 31, 84, 105, 38, 178, 104, 202, 88,
218, 62, 10, 80, 77, 37, 162, 190, 30, 16, 5, 54, 44,
110, 262, 107, 134, 103, 96, 57, 240, 94, 113, 56, 64, 320,
 9, 129, 58, 126, 166, 95, 97, 204, 116, 161, 146, 302, 53,
143, 124, 214, 288, 306, 322, 67, 61, 130, 220, 78, 314, 226,
280, 91, 234, 163, 151, 85, 256, 168, 144, 66, 128, 73, 122,
154, 63, 292, 188, 71, 135, 138, 184, 89, 157, 118, 111, 192,
127, 216, 139, 196, 99, 167, 276, 121, 69, 155, 316, 242, 304,
284, 278, 310, 222, 212, 250, 180, 258, 330, 158, 149, 165, 79,
194, 176, 174, 312, 200, 328, 300, 153, 232, 336, 308, 147, 298,
224, 254, 248, 236, 170, 264, 119, 117, 334, 324, 1, 238, 266,
282, 268, 244, 272, 294, 270, 286], dtype=int64)

```
In [28]: df["training_hours"].isna().sum()
```

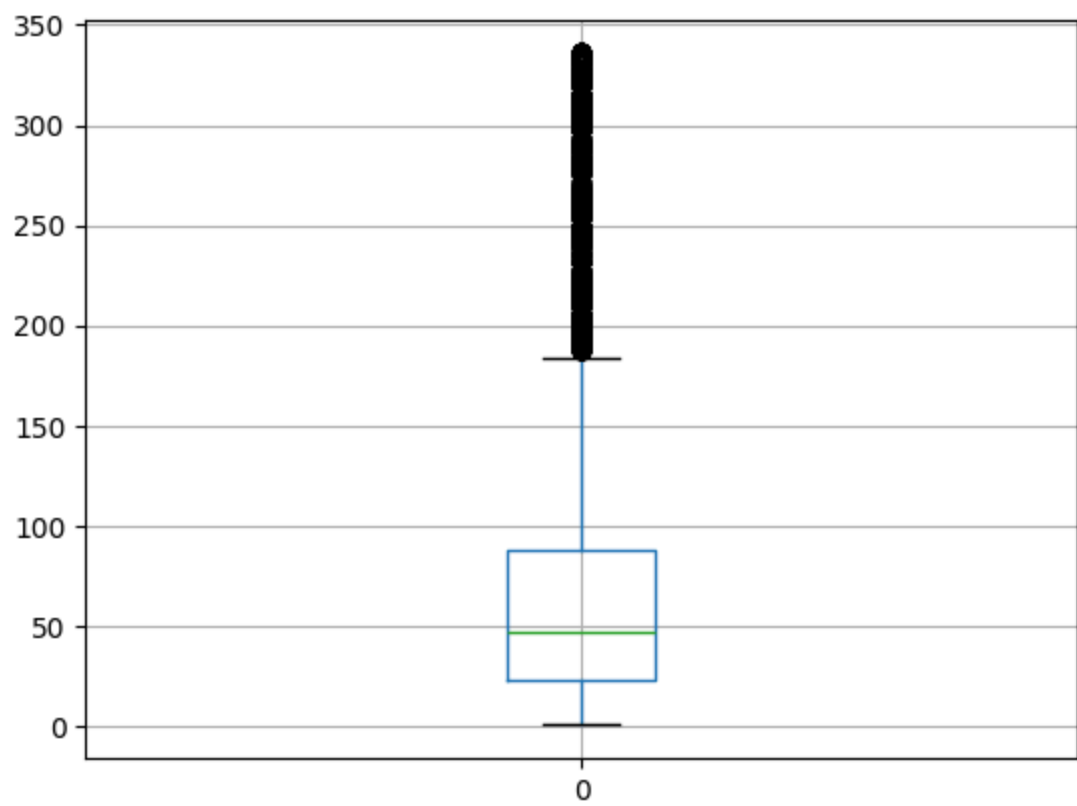
Out[28]: 0

Outlier Analysis

```
In [29]: sorted = list(df["training_hours"])
sorted.sort()
```

```
In [ ]:
```

```
In [30]: pd.DataFrame(sorted).boxplot()
plt.show()
```



```
In [31]: q1_term = 0.25*len(sorted)+1
q3_term = 0.75*len(sorted)+1
q1_term,q3_term
```

Out[31]: (4790.5, 14369.5)

```
In [32]: q1_decimal = q1_term - int(q1_term)
q1_decimal
```

Out[32]: 0.5

```
In [33]: q1_part1 = sorted[int(q1_term)+1]
q1_part1
```

Out[33]: 23

```
In [34]: q1_part2 = q1_decimal*(sorted[int(q1_term)+1+1] - sorted[int(q1_term)+1])
q1_part2
```

Out[34]: 0.0

```
In [35]: q1 = q1_part1 + q1_part2
q1
```

Out[35]: 23.0

```
In [36]: q3_decimal = q3_term - int(q3_term)
q3_decimal
```

Out[36]: 0.5

```
In [37]: q3_part1 = sorted[int(q3_term)+1]
q3_part1
```

Out[37]: 88

```
In [38]: q3_part2 = q3_decimal*(sorted[int(q3_term)+1+1] - sorted[int(q3_term)+1])
q3_part2
```

Out[38]: 0.0

```
In [39]: q3 = q3_part1 + q3_part2
q3
```

Out[39]: 88.0

```
In [40]: iqr = q3-q1
iqr
```

Out[40]: 65.0

```
In [41]: min = q1 - iqr
min
```

Out[41]: -42.0

```
In [42]: max = q3+iqr
max
```

Out[42]: 153.0

```
In [43]: ##so outliers are at the upper end
```

```
In [44]: df["training_hours"].describe()
```

Out[44]: count 19158.000000
mean 65.366896
std 60.058462
min 1.000000
25% 23.000000
50% 47.000000
75% 88.000000
max 336.000000
Name: training_hours, dtype: float64

```
In [45]: ###95th percentile
q95_term = 0.92*len(sorted)+1
q95_term
```

Out[45]: 17626.36

```
In [46]: q95_decimal = q95_term - int(q95_term)
q95_decimal
```

Out[46]: 0.36000000000005821

```
In [47]: q95_part1 = sorted[int(q95_term)+1]
q95_part1
```

Out[47]: 158

```
In [48]: q95_part2 = q95_decimal*(sorted[int(q95_term)+1+1] - sorted[int(q95_term)+1])
q95_part2
```

Out[48]: 0.0

```
In [49]: q95 = q95_part1 + q95_part2
q95
```

Out[49]: 158.0

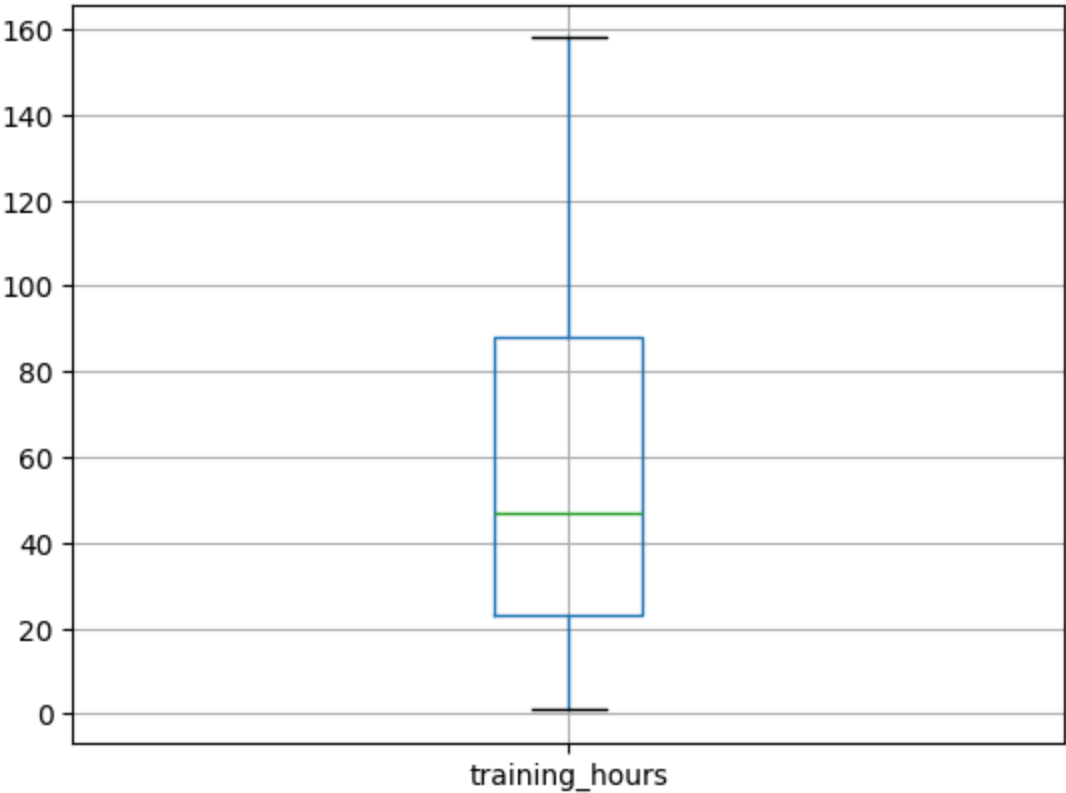
```
In [50]: df["training_hours"][df["training_hours"]>=q95]
```

```
Out[50]: 35      206
        69      228
        89      332
        91      182
        93      172
        ...
       19107    322
       19114    160
       19117    214
       19118    316
       19145    190
Name: training_hours, Length: 1546, dtype: int64
```

```
In [51]: df.loc[df['training_hours'] >= max, 'training_hours'] = q95
df["training_hours"].max()
```

Out[51]: 158

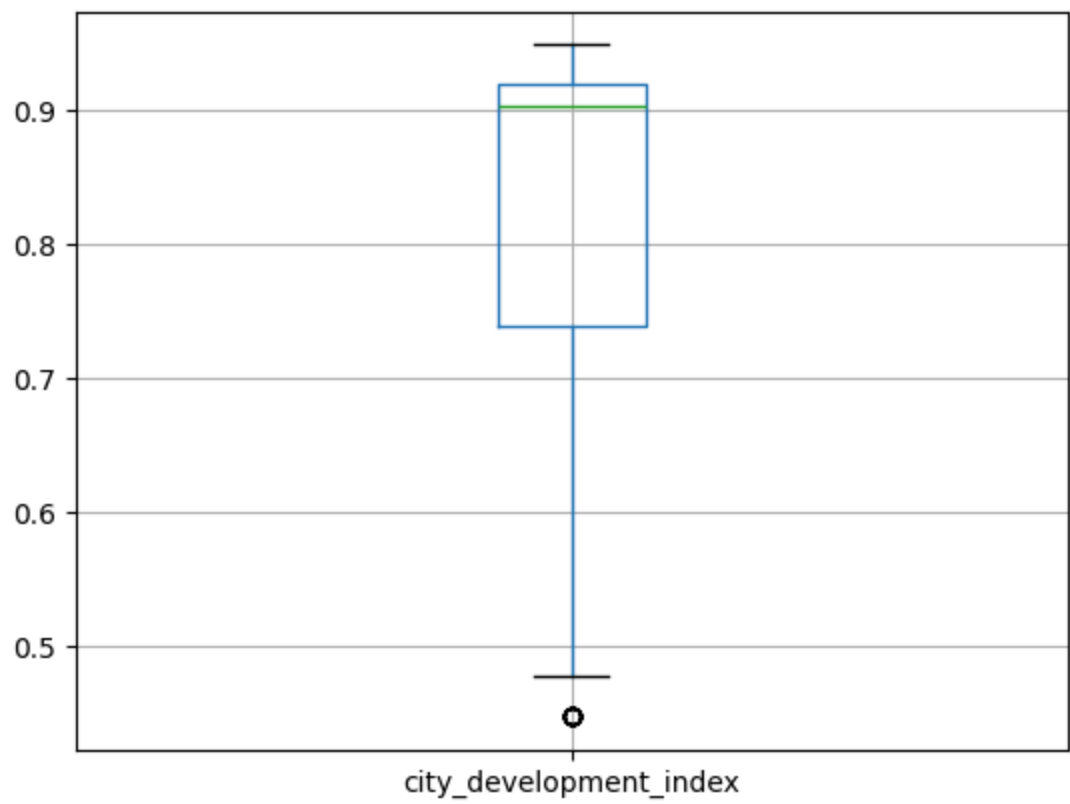
```
In [52]: pd.DataFrame(df["training_hours"]).boxplot()
plt.show()
```



```
In [53]: df["training_hours"].skew()
```

Out[53]: 0.8404856267981223

```
In [54]: pd.DataFrame(df["city_development_index"]).boxplot()
plt.show()
```

```
In [55]: iqr = df["city_development_index"].describe()["75%"]-df["city_development_index"].describe()["25%"]
iqr
```

Out[55]: 0.18000000000000005

```
In [56]: min = df["city_development_index"].describe()["25%"]-iqr
max = df["city_development_index"].describe()["75%"]+iqr
```

```
In [57]: min,max
```

Out[57]: (0.5599999999999999, 1.1)

```
In [58]: df["city_development_index"][df["city_development_index"]<min].count()
```

Out[58]: 577

```
In [59]: df["city_development_index"][df["city_development_index"]>max].count()
```

Out[59]: 0

```
In [60]: sorted = list(df["city_development_index"])
sorted.sort()
sorted
```


[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
In [61]: q_5 = 0.05*len(sorted)+1
          q_95 = 0.95*len(sorted)+1
          q_5, q_95
```

```
Out[61]: (958.9000000000001, 18201.1)
```

```
In [62]: decimal = q_5 - int(q_5)
          decimal
```

Out[62]: 0.9000000000000091

```
In [63]: q_5_part1 = sorted[int(q_5)+1]
          q_5_part1
```

Out[63]: 0.624

```
In [64]: q_5_part2 = decimal*(sorted[int(q_5_part1)+1+1]-sorted[int(q_5_part1)+1])
q_5_part2
```

Out[64]: 0.0

```
In [65]: q_5val = q_5_part1+q_5_part2
q_5val
```

Out[65]: 0.624

```
In [66]: decimal = q_95 - int(q_95)
decimal
```

Out[66]: 0.09999999999854481

```
In [67]: q_95_part1 = sorted[int(q_95)+1]
q_95_part1
```

Out[67]: 0.926

```
In [68]: q_95_part2 = decimal*(sorted[int(q_95_part1)+1+1]-sorted[int(q_95_part1)+1])
q_95_part2
```

Out[68]: 0.0

```
In [69]: q_95val = q_95_part1+q_95_part2
q_95val
```

Out[69]: 0.926

```
In [70]: df.loc[df['city_development_index'] <= min, 'city_development_index'] = q_5
df.loc[df['city_development_index'] >= max, 'city_development_index'] = q_95
df["city_development_index"].unique()
```

Out[70]: array([9.20000e-01, 7.76000e-01, 6.24000e-01, 7.89000e-01, 7.67000e-01,
7.64000e-01, 7.62000e-01, 9.13000e-01, 9.26000e-01, 8.27000e-01,
8.43000e-01, 8.04000e-01, 8.55000e-01, 8.87000e-01, 9.10000e-01,
8.84000e-01, 9.24000e-01, 6.66000e-01, 1.82011e+04, 9.23000e-01,
7.94000e-01, 7.54000e-01, 9.39000e-01, 8.65000e-01, 6.98000e-01,
8.93000e-01, 7.96000e-01, 8.66000e-01, 6.82000e-01, 8.02000e-01,
5.79000e-01, 8.78000e-01, 8.97000e-01, 9.49000e-01, 9.25000e-01,
8.96000e-01, 8.36000e-01, 6.93000e-01, 7.69000e-01, 7.75000e-01,
9.03000e-01, 7.27000e-01, 6.40000e-01, 7.43000e-01, 8.99000e-01,
9.15000e-01, 6.89000e-01, 8.95000e-01, 8.90000e-01, 8.47000e-01,
7.66000e-01, 7.38000e-01, 6.47000e-01, 7.95000e-01, 7.40000e-01,
7.01000e-01, 8.40000e-01, 6.91000e-01, 7.35000e-01, 7.42000e-01,
7.22000e-01, 9.21000e-01, 8.48000e-01, 8.56000e-01, 8.98000e-01,
8.30000e-01, 7.30000e-01, 6.80000e-01, 7.25000e-01, 7.63000e-01,
7.45000e-01, 6.45000e-01, 7.88000e-01, 7.80000e-01, 7.39000e-01,
5.63000e-01, 8.24000e-01, 6.49000e-01, 7.81000e-01, 6.25000e-01,
8.07000e-01, 6.64000e-01])

```
In [71]: pd.DataFrame(df["city_development_index"]).boxplot()
plt.show()
```




```
In [72]: df.head()
```

Out[72]:

	enrollee_id	city_development_index	gender	relevent_experience	enrolled_university	ec
0	8949	0.920	Male	Has relevent experience	no_enrollment	
1	29725	0.776	Male	No relevent experience	no_enrollment	
2	11561	0.624	Male	No relevent experience	Full time course	
3	33241	0.789	Male	No relevent experience	no_enrollment	
4	666	0.767	Male	Has relevent experience	no_enrollment	

```
In [73]: df["gender"].unique()
```

Out[73]: array(['Male', 'Female', 'Other'], dtype=object)

```
In [74]: df = pd.get_dummies(df,columns=["gender","major_discipline","company_type"],drop_fi
df.head()
```

Out[74]:

	enrollee_id	city_development_index	relevent_experience	enrolled_university	education_l
0	8949	0.920	Has relevent experience	no_enrollment	Grad
1	29725	0.776	No relevent experience	no_enrollment	Grad
2	11561	0.624	No relevent experience	Full time course	Grad
3	33241	0.789	No relevent experience	no_enrollment	Grad
4	666	0.767	Has relevent experience	no_enrollment	Ma

5 rows × 25 columns



In [75]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19158 entries, 0 to 19157
Data columns (total 25 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   enrollee_id                          19158 non-null  int64
1   city_development_index               19158 non-null  float64
2   relevent_experience                  19158 non-null  object
3   enrolled_university                 19158 non-null  object
4   education_level                      19158 non-null  object
5   experience                           19158 non-null  float64
6   company_size                         19158 non-null  object
7   last_new_job                         19158 non-null  float64
8   training_hours                      19158 non-null  int64
9   target                              19158 non-null  float64
10  gender_Female                        19158 non-null  int32
11  gender_Male                          19158 non-null  int32
12  gender_Other                         19158 non-null  int32
13  major_discipline_Arts                19158 non-null  int32
14  major_discipline_Business Degree    19158 non-null  int32
15  major_discipline_Humanities          19158 non-null  int32
16  major_discipline_No Major            19158 non-null  int32
17  major_discipline_Other               19158 non-null  int32
18  major_discipline_STEM                19158 non-null  int32
19  company_type_Early Stage Startup     19158 non-null  int32
20  company_type_Funded Startup          19158 non-null  int32
21  company_type_NGO                     19158 non-null  int32
22  company_type_Other                   19158 non-null  int32
23  company_type_Public Sector            19158 non-null  int32
24  company_type_Pvt Ltd                  19158 non-null  int32
dtypes: float64(4), int32(15), int64(2), object(4)
memory usage: 2.6+ MB
```

In [76]:

```
values = df["company_size"].unique()
values
```

Out[76]:

```
array(['50-99', '0-10', '10000+', '5000-9999', '1000-4999', '10-49',
       '100-500', '500-999'], dtype=object)
```

In [77]:

```
labels = [2,0,7,6,5,1,3,4]
values,labels
```

Out[77]:

```
(array(['50-99', '0-10', '10000+', '5000-9999', '1000-4999', '10-49',
       '100-500', '500-999'], dtype=object),
 [2, 0, 7, 6, 5, 1, 3, 4])
```

In [78]:

```
df["company_size"] = df["company_size"].replace(values,labels)
df["company_size"].unique()
```

```
C:\Users\Gaurish Garg\AppData\Local\Temp\ipykernel_76520\2256779259.py:1: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version. To retain the old behavior, explicitly call `result.infer_objects(copy=False)`. To opt-in to the future behavior, set `pd.set_option('future.no_silent_downcasting', True)`
df["company_size"] = df["company_size"].replace(values,labels)
```

Out[78]: array([2, 0, 7, 6, 5, 1, 3, 4], dtype=int64)

```
In [79]: values = df["relevent_experience"].unique()
```

```
In [80]: labels = [1,0]
         values,labels
```

Out[80]: (array(['Has relevent experience', 'No relevent experience'], dtype=object), [1, 0])

```
In [81]: df["relevent_experience"] = df["relevent_experience"].replace(values,labels)
         df["relevent_experience"].unique()
```

```
C:\Users\Gaurish Garg\AppData\Local\Temp\ipykernel_76520\1211771326.py:1: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version. To retain the old behavior, explicitly call `result.infer_objects(copy=False)`. To opt-in to the future behavior, set `pd.set_option('future.no_silent_downcasting', True)`
df["relevent_experience"] = df["relevent_experience"].replace(values,labels)
```

Out[81]: array([1, 0], dtype=int64)

```
In [82]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19158 entries, 0 to 19157
Data columns (total 25 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   enrollee_id                          19158 non-null  int64
1   city_development_index               19158 non-null  float64
2   relevent_experience                  19158 non-null  int64
3   enrolled_university                 19158 non-null  object
4   education_level                     19158 non-null  object
5   experience                           19158 non-null  float64
6   company_size                         19158 non-null  int64
7   last_new_job                         19158 non-null  float64
8   training_hours                      19158 non-null  int64
9   target                              19158 non-null  float64
10  gender_Female                       19158 non-null  int32
11  gender_Male                         19158 non-null  int32
12  gender_Other                        19158 non-null  int32
13  major_discipline_Arts               19158 non-null  int32
14  major_discipline_Business Degree    19158 non-null  int32
15  major_discipline_Humanities         19158 non-null  int32
16  major_discipline_No Major           19158 non-null  int32
17  major_discipline_Other              19158 non-null  int32
18  major_discipline_STEM               19158 non-null  int32
19  company_type_Early Stage Startup    19158 non-null  int32
20  company_type_Funded Startup         19158 non-null  int32
21  company_type_NGO                    19158 non-null  int32
22  company_type_Other                  19158 non-null  int32
23  company_type_Public Sector          19158 non-null  int32
24  company_type_Pvt Ltd                19158 non-null  int32
dtypes: float64(4), int32(15), int64(4), object(2)
memory usage: 2.6+ MB
```

```
In [83]: values = df["enrolled_university"].unique()
         values
```

Out[83]: array(['no_enrollment', 'Full time course', 'Part time course'], dtype=object)

```
In [84]: labels = [0,2,1]
```

```
In [85]: df["enrolleed_university"] = df["enrolled_university"].replace(values,labels)
df["enrolleed_university"].unique()
```

C:\Users\Gaurish Garg\AppData\Local\Temp\ipykernel_76520\2300801064.py:1: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version. To retain the old behavior, explicitly call `result.infer\_objects(copy=False)`. To opt-in to the future behavior, set `pd.set\_option('future.no\_silent\_downcasting', True)`

```
df["enrolleed_university"] = df["enrolled_university"].replace(values,labels)
```

Out[85]: array([0, 2, 1], dtype=int64)

```
In [86]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19158 entries, 0 to 19157
Data columns (total 25 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   enrollee_id                               19158 non-null  int64
1   city_development_index                    19158 non-null  float64
2   relevent_experience                       19158 non-null  int64
3   enrolled_university                     19158 non-null  int64
4   education_level                           19158 non-null  object
5   experience                               19158 non-null  float64
6   company_size                             19158 non-null  int64
7   last_new_job                             19158 non-null  float64
8   training_hours                           19158 non-null  int64
9   target                                   19158 non-null  float64
10  gender_Female                             19158 non-null  int32
11  gender_Male                               19158 non-null  int32
12  gender_Other                             19158 non-null  int32
13  major_discipline_Arts                    19158 non-null  int32
14  major_discipline_Business Degree         19158 non-null  int32
15  major_discipline_Humanities              19158 non-null  int32
16  major_discipline_No Major                19158 non-null  int32
17  major_discipline_Other                   19158 non-null  int32
18  major_discipline_STEM                    19158 non-null  int32
19  company_type_Early Stage Startup          19158 non-null  int32
20  company_type_Funded Startup              19158 non-null  int32
21  company_type_NGO                         19158 non-null  int32
22  company_type_Other                       19158 non-null  int32
23  company_type_Public Sector                19158 non-null  int32
24  company_type_Pvt Ltd                     19158 non-null  int32
dtypes: float64(4), int32(15), int64(5), object(1)
memory usage: 2.6+ MB
```

```
In [87]: df.head()
```

Out[87]:

	enrollee_id	city_development_index	relevent_experience	enrolled_university	education_l
0	8949	0.920	1	0	Grad
1	29725	0.776	0	0	Grad
2	11561	0.624	0	2	Grad
3	33241	0.789	0	0	Grad
4	666	0.767	1	0	Ma

5 rows × 25 columns

```
In [88]: values = df["education_level"].unique()
```

```
In [89]: values
```

Out[89]: array(['Graduate', 'Masters', 'High School', 'Phd', 'Primary School'], dtype=object)

```
In [90]: labels = [2,3,1,4,0]
labels
```

Out[90]: [2, 3, 1, 4, 0]

```
In [91]: df["education_level"] = df["education_level"].replace(values,labels)
df["education_level"].unique()
```

C:\Users\Gaurish Garg\AppData\Local\Temp\ipykernel_76520\3766738430.py:1: FutureWarning: Downcasting behavior in `replace` is deprecated and will be removed in a future version. To retain the old behavior, explicitly call `result.infer\_objects(copy=False)`. To opt-in to the future behavior, set `pd.set\_option('future.no\_silent\_downcasting', True)`

```
df["education_level"] = df["education_level"].replace(values,labels)
```

Out[91]: array([2, 3, 1, 4, 0], dtype=int64)

```
In [92]: df.head()
```

Out[92]:

	enrollee_id	city_development_index	relevent_experience	enrolled_university	education_l
0	8949	0.920	1	0	
1	29725	0.776	0	0	
2	11561	0.624	0	2	
3	33241	0.789	0	0	
4	666	0.767	1	0	

5 rows × 25 columns

```
In [93]: columns = list(df.columns)
columns
```

Out[93]: ['enrollee_id',
'city_development_index',
'relevent_experience',
'enrolled_university',
'education_level',
'experience',
'company_size',
'last_new_job',
'training_hours',
'target',
'gender_Female',
'gender_Male',
'gender_Other',
'major_discipline_Arts',
'major_discipline_Business Degree',
'major_discipline_Humanities',
'major_discipline_No Major',
'major_discipline_Other',
'major_discipline_STEM',
'company_type_Early Stage Startup',
'company_type_Funded Startup',
'company_type_NGO',
'company_type_Other',
'company_type_Public Sector',
'company_type_Pvt Ltd']

```
In [94]: columns.remove("enrollee_id")
columns
```

```
Out[94]: ['city_development_index',
          'relevent_experience',
          'enrolled_university',
          'education_level',
          'experience',
          'company_size',
          'last_new_job',
          'training_hours',
          'target',
          'gender_Female',
          'gender_Male',
          'gender_Other',
          'major_discipline_Arts',
          'major_discipline_Business Degree',
          'major_discipline_Humanities',
          'major_discipline_No Major',
          'major_discipline_Other',
          'major_discipline_STEM',
          'company_type_Early Stage Startup',
          'company_type_Funded Startup',
          'company_type_NGO',
          'company_type_Other',
          'company_type_Public Sector',
          'company_type_Pvt Ltd']
```

```
In [95]: columns.remove("target")
```

```
In [96]: x = df.loc[:,columns]
          y = df.loc[:, "target"]
```

```
In [97]: x
```

Out[97]:

	city_development_index	relevent_experience	enrolled_university	education_level	e
0	0.920	1	0	2	
1	0.776	0	0	2	
2	0.624	0	2	2	
3	0.789	0	0	2	
4	0.767	1	0	3	
...	...	...	...	...	...
19153	0.878	0	0	2	
19154	0.920	1	0	2	
19155	0.920	1	0	2	
19156	0.802	1	0	1	
19157	0.855	0	0	0	

19158 rows × 23 columns

```
In [98]: y
```

```
Out[98]: 0      1.0
         1      0.0
         2      0.0
         3      1.0
         4      0.0
         ...
        19153    1.0
        19154    1.0
        19155    0.0
        19156    0.0
        19157    0.0
        Name: target, Length: 19158, dtype: float64
```

```
In [99]: columns = x.columns
         columns
```

```
Out[99]: Index(['city_development_index', 'relevent_experience', 'enrolled_university',
               'education_level', 'experience', 'company_size', 'last_new_job',
               'training_hours', 'gender_Female', 'gender_Male', 'gender_Other',
               'major_discipline_Arts', 'major_discipline_Business Degree',
               'major_discipline_Humanities', 'major_discipline_No Major',
               'major_discipline_Other', 'major_discipline_STEM',
               'company_type_Early Stage Startup', 'company_type_Funded Startup',
               'company_type_NGO', 'company_type_Other', 'company_type_Public Sector',
               'company_type_Pvt Ltd'],
              dtype='object')
```

```
In [ ]:
```

```
In [100... for column in columns:
            min = x[column].min()
            max = x[column].max()
            den = max-min
            x[column] = (x[column]-min)/den
            x.head()
```

Out[100...

	city_development_index	relevent_experience	enrolled_university	education_level	experience
0	0.000020	1.0	0.0	0.50	1.00
1	0.000012	0.0	0.0	0.50	0.71
2	0.000003	0.0	1.0	0.50	0.23
3	0.000012	0.0	0.0	0.50	0.00
4	0.000011	1.0	0.0	0.75	1.00

5 rows × 23 columns



```
In [101... x.to_csv("preprocessed_Data.csv")
            y.to_csv("target.csv")
```

```
In [103... import matplotlib.pyplot as plt
import numpy as np

corr = x.corr()

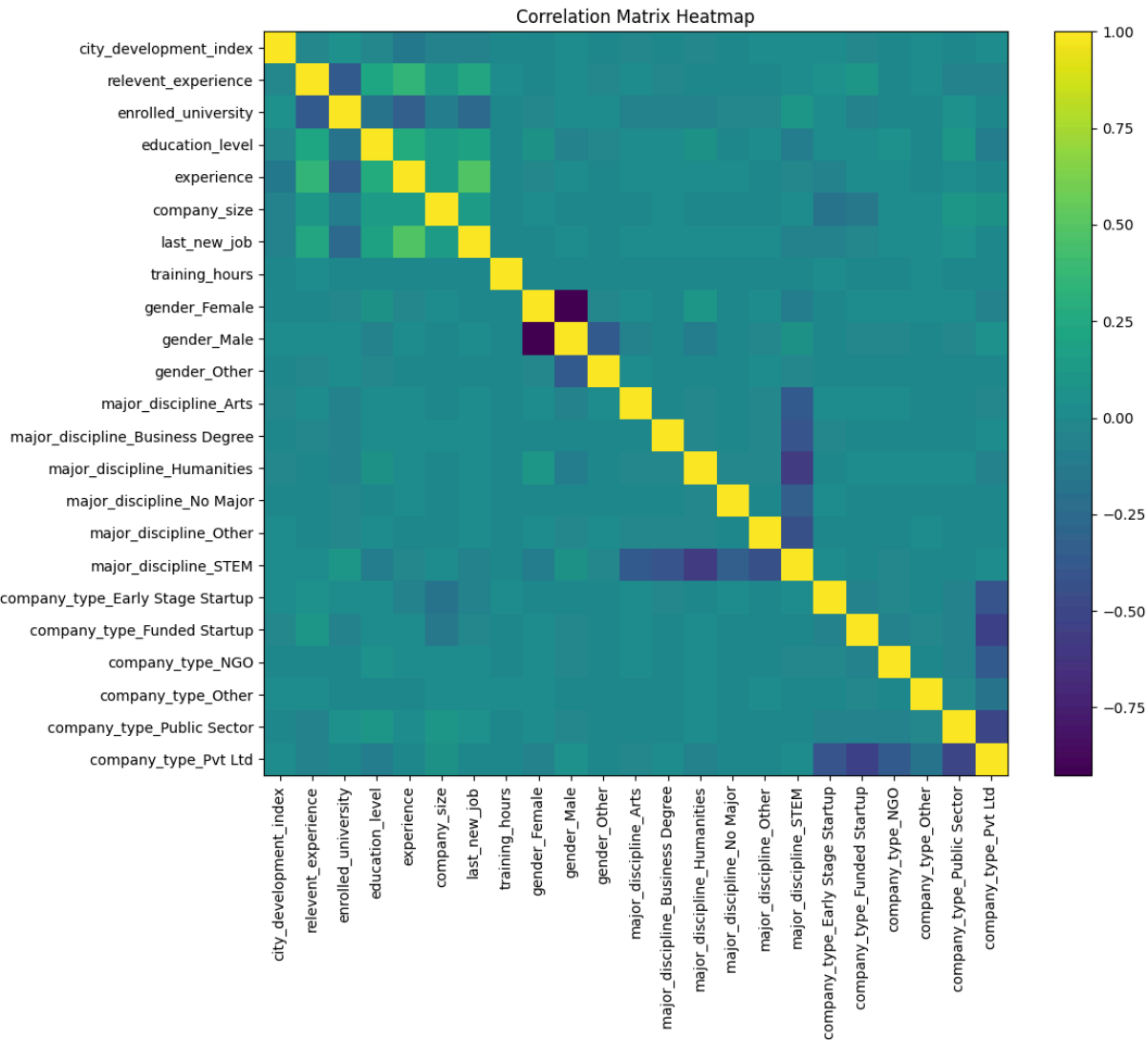
plt.figure(figsize=(12, 10))

plt.imshow(corr, interpolation='nearest')
plt.colorbar()

# Add Labels
plt.xticks(ticks=np.arange(len(corr.columns)), labels=corr.columns, rotation=90)
plt.yticks(ticks=np.arange(len(corr.columns)), labels=corr.columns)

plt.title("Correlation Matrix Heatmap")
```

```
plt.tight_layout()  
plt.show()
```



```
In [ ]:  
  
In [ ]:  
  
In [ ]:
```